H-B-1490A/GEN
October 17, 1973
SUPERSEDING
Fed. Spec. H-B-1490
September 18, 1969
(See 6.3)

FEDERAL SPECIFICATION

BRUSHES, SCRUB

GENERAL SPECIFICATION FOR

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers the general requirements for scrub brushes. Specific requirements for scrub brushes are covered by the applicable specification sheets (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-S-951	-	Synthetic Filaments For Brushes; General Specification For
L-P-394	-	Plastic Molding Material. (Propylene Plastics, Injection and Extrusion)
L-P-398	-	Plastic Molding Material, Styrene-Butadiene
PPP-B-1729	-	Brushes, Packaging and Packing Of

Federal Standard:

FED-STD-66	-	Steel: Chemical Composition and Hardenability

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-M-20693 - Molding Plastic, Polyamide (Nylon), Rigid

See Supplement-1 for list of associated specification sheets.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1363 - Measurement of Wood Moisture Content

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Standard product. The scrub brushes delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract shall be new and interchangeable.

3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial scrub brush provided the completed item complies with all provisions of this specification.

3.3.1 Wood. Wood shall be clear, close-grained and straight-grained hardwood. At the time of assembly, the wood shall have a moisture content of not more than 12 percent when tested as specified in 4.1.1.1.

3.3.2 Plastic molding. Plastic molding material shall conform to the applicable requirements hereinafter specified. Clean, unburned plastic materials in the form of imperfect parts, sprues, runners, or other scrap of the same composition as virgin material and produced in the molding or finishing operations, may be reground and mixed with virgin material.

3.3.2.1 Modified general purpose polypropylene. The molding material shall by type I polypropylene of L-P-394 except that the physical requirements shall be limited to those given in table I when tested as specified in 4.1.1.2.

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TABLE I. Requirements for polypropylene plastic

Property at 23°C (73°F)	Units	Property values
Density	gm/ml	Min 0.901 Max 0.907
Tensile yield strength	psi	Min 4900
Izod impact strength	ft lb/in.	Min 0.5
Rockwell hardness	R scale	Min 96

3.3.2.2 General purpose or impact resistant polypropylene. The general purpose molding material shall conform to type I of L-P-394. The impact resistant molding material shall conform to type III, grade A or B of L-P-394.

3.3.2.3 High impact resistant polypropylene. The molding material shall conform to type III, grade B of L-P-394.

3.3.2.4 Styrene-Butadiene. The molding material shall conform to type IV, class 2 of L-P-398.

3.3.2.5 Nylon. The molding material shall conform to white color, composition A, type I of MIL-M-20693.

3.3.3 Filler.

3.3.3.1 Polyamide. Polyamide filler material shall conform to class N of H-S-951 and in accordance with specification sheet requirements.

3.3.3.2 Styrene. Styrene filler material shall be 0.016 inch diameter and shall conform to class S, type III of H-S-951.

3.3.3.3 Polypropylene. Polypropylene filler material shall conform to class P, type II of H-S-951.

3.3.3.4 Tampico. Fiber shall be new and unused commercially available tula tampico fiber.

3.3.3.5 Palmetto. Fiber shall be new and unused commercially available oiled palmetto fiber.

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3.3.3.6 Palmyra. Fiber shall be straight, resilient, no artificial color added, new and unused commercially available palmyra fiber.

3.3.4 Staple wire.

3.3.4.1 Corrosion resisting steel (CRES). CRES staple wire shall be commercial quality and not less than 0.041 inch diameter.

3.3.4.2 Carbon steel. Carbon steel staple wire shall be commercial quality, not less than 0.041 or 0.047 inch diameter in accordance with specification sheet requirements, and finished to resist corrosion.

3.3.4.3 Type 302 corrosion resisting steel (CRES). Type 302 (CRES) staple wire shall be round or flat and conform to FED-STD-66. The round wire shall be not less than 0.020 inch diameter. The flat wire shall be not less than 0.000314 square inch in cross-sectional area.

3.4 Design. Except as otherwise noted thereon, the illustrations on the specification sheets are intended for information purposes only and are not intended to preclude brushes otherwise meeting the requirements of this specification and applicable specification sheets.

3.5 Construction. Brushes shall have staple set filler materials and shall be fabricated as specified herein and in applicable specification sheets. The brush filler material lengths shall be evenly and uniformly trimmed to conform to the dimensional requirements of the finished brushes as specified in the applicable specification sheets. The weight of the filler material, when applicable, shall be not less than as specified in the applicable specification sheet when tested as specified in 4.5.1. Staples and filler material shall be driven to the bottom of the tuft holes and the staples imbedded in the block at the bottom of the holes. The set staples shall be capable of withstanding the retention test specified in 4.5.2.1. Individual strands of filler material shall resist removal when tested as specified in 4.5.2.2. When specified on the specification sheet, brushes shall be capable of passing the heat resistance test in 4.5.3 without evidence of permenent distortion, cracking, loss of bristles, or discoloration. The palmyra fiber shall be capable of passing the soak test without any evidence of artificial color of fiber or any perceptible change in stiffness of fiber when tested as specified in 4.5.4. The carbon steel staple wire of the applicable scrub brushes shall show no evidence of corrosion when tested as specified in 4.5.5.

3.5.1 Blocks. Blocks shall be of one-piece construction. All edges of the blocks shall be rounded. Plastic molded blocks shall be provided with a scored, corrugated, or slip resistant gripping surface. Wood blocks shall be smoothly finished, i.e., free of roughness. Tuft holes in the brush blocks shall be uniformly distributed in the quantity noted on the applicable specification sheet.

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3.5.2 Identification marking. Each brush block shall be permanently and legibly indented or marked with the manufacturer's name or trade name, or trademark of such characters as to be identifiable with the manufacturer.

3.5.3 Finish. Brush blocks shall be finished in accordance with the manufacturer's current commercial practice.

3.6 Workmanship. The finished brushes shall be clean and free from oil, grease, and dirt. Blocks shall be free of any crack, rough or sharp edges or corners. Wood blocks shall be free of loose grain and splinters. Filler material shall be free of foreign and loose materials. The legs of the staples shall not penetrate through the block.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.1.1.1 Moisture content of wood blocks. When wood blocks are furnished, the moisture content of wood requirement as specified in 3.3.1 shall be determined in accordance with MIL-STD-1363 at time of assembly. The lot shall be all wood blocks of one type and size prepared under the same conditions, stored as a group, and submitted for inspection at one time. The sample unit shall be one block. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units. Three determinations shall be made on each sample unit. The average of the three determinations shall be the moisture content of the sample unit. Any sample unit of wood having more than 12 percent moisture content shall be classified as a defect.

4.1.1.2 Modified general purpose polypropylene molding material. When modified general purpose polypropylene blocks are furnished, testing of the

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blocks shall be in accordance with L-P-394 and procedure A of ASTM Test Method D 785. Three determinations of hardness shall be made on each sample unit. The average of the three determinations to the nearest Rockwell scale reading shall be the Rockwell hardness of the sample unit. The lot shall be all like modified general purpose polypropylene blocks submitted for inspection at one time. The sample unit shall be one block. The inspection level shall be S-1. The lot shall be unacceptable if one or more samples fails to comply with all property values in table I (see 3.3.2.1).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3)
(b) Inspection of preparation for delivery (see 4.4)

4.3 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.3.1 End item inspection. The inspection lot shall be all like brushes submitted for inspection at one time. The sample unit shall be one brush.

4.3.1.1 Visual examination. The brushes shall be examined for the defects in table II. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Wood block not smoothly finished, i.e., is rough		X
	Finish is not manufacturer's current commercial practice		X
Construction	Hardwood block is not clear, close-grained or straight grained	X	
	Brush filler material lengths not evenly or uniformly trimmed		X
	Blocks are more than one piece construction	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Construction (cont'd)	Any edge of block not rounded		X
	Block is not provided with a shaped, scored, corrugated, or slip resistant gripping surface or handle as applicable		X
	Surgical scrub brush block does not have vertically serrated sides properly shaped to provide a firm grip		X
	Hang-up hole not provided in brush when required		X
	Tuft holes in the block not located to insure an even distribution of filler material		X
Marking for identification	Missing, illegible, or incomplete		X
Workmanship	Not clean, i.e., evidence of oil, grease or dirt		X
	Any block having crack, rough or sharp edge or corner	X	
	Wood block has loose grain or splinter		X
	Foreign or loose filler material	X	
	Legs of staple penetrate through the block	X	

4.3.1.2 Dimensional examination. The brushes shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

4.3.1.3 End item testing. Testing of the completely fabricated brushes shall be performed in accordance with table III for the characteristics shown therein. The inspection lot shall consist of all like brushes submitted for inspection at one time. The sample unit shall be one brush. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

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TABLE III. Testing of the end item

Characteristic	Requirement reference	Test method reference	Results reported as
Weight of filler material; when applicable	3.5	4.5.1	Pass or fail
Setting of staples	3.5	4.5.2.1	Pass or fail
Set of filler material	3.5	4.5.2.2	Pass or fail
Heat resistance, when applicable	3.5	4.5.3	Pass or fail
Color and resilienc of palmyra fiber	3.5	4.5.4	Pass or fail
Corrosion resistance of carbon steel staple wire	3.5	4.5.5	Pass or fail

4.4 Examination of preparation for delivery. An examination shall be made in accordance with PPP-B-1729 to determine compliance with section 5.

4.5 Tests.

4.5.1 Weight of filler material.

4.5.1.1 Polyamide, styrene, and polypropylene. The unconditioned polyamide, styrene, and polypropylene filler material, as applicable, shall be removed from the brush and weighed. Any weight less than the specified requirement shall be classified as a defect (see 3.5).

4.5.1.2 Tampico, palmetto, and palmyra. Tampico, palmetto and palmyra filler material, as applicable, shall be removed from the brush and conditioned for a period of not less than 16 hours in air at a relative humidity of 50 percent plus or minus 4 percent and at a temperature of 23 degrees plus or minus 2 degrees centigrade. Following the conditioning period, the filler material shall be weighed. Any weight less than the specified requirement shall be classified as a defect (see 3.5).

4.5.2 Set retention.

4.5.2.1 Set retention of staples. The block shall be held with a holding device. Individual tufts of filler material shall be grasped with another

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holding device and, with a spring scale or weight, a pulling force of eight pounds shall be applied in a direction tending to pull the tuft and the staple from the block. Determinations shall be made on ten different tufts on each brush. Any staple pulled loose from the block as the result of the test shall be classified as a defect (see 3.5).

4.5.2.2 Set retention of filler material. A single strand of filler material from ten random tufts, in turn, shall be firmly grasped between the thumb and forefinger and pulled in an attempt to remove it. Any strand removed before breaking or slipping from the grasp shall be classified as a defect (see 3.5).

4.5.3 Heat resistance test. Brushes shall be autoclaved at 250 degrees F. for a period of 30 minutes. After the autoclaving period, the brushes shall be removed and allowed to dry at room temperature for a period of one hour. The 30 minute period of autoclaving at 250 degrees F. followed by the drying period of one hour shall be repeated four times. The brushes shall then be autoclaved at 270 degrees F. for a period of 10 minutes followed by the drying period of one hour. The 10 minute period of autoclaving at 270 degrees F. followed by the drying period of one hour shall be repeated four times. The brushes shall then be examined for compliance with 3.5. Any brush having evidence of permanent distortion, cracking, discoloration, or loss of bristles shall be classified as a defect.

4.5.4 Soak test. Palmyra fiber shall be soaked in boiling water (212 degrees F.) for a period of not less than one hour. Following the hour soaking period, the palmyra fiber shall be examined for compliance with the applicable requirements under 3.5. Any brush having evidence of artificial color of fiber or any perceptible change in stiffness of fiber shall be classified as a defect.

4.5.5 Corrosion resistance test. Carbon steel staple wire shall be removed from the sample units. The wire staples shall be placed in a boiling 10 percent (by weight) aqueous solution of sodium chloride for a period of not less than 15 minutes. Following the 15 minute period, the wire staples shall be removed from the solution and immediately immersed in 10 percent (by weight) aqueous solution of sodium chloride at room temperature for a period of not less than one hour. Following the one hour period, the wire staples shall be removed from the solution and, without having the adhering liquid wiped off, shall be allowed to dry at room temperature for a period of not less than 24 hours. Following the 24 hour period, the wire staples shall be examined for compliance with 3.5. Any brush having staple wire with evidence of corrosion shall be classified as a defect.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, marking and palletization shall be in accordance with PPP-B-1729 for level A, B, or C as specified (see 6.2).

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6. NOTES

6.1 Intended use. The scrub brushes are intended to be used for the cleaning purposes indicated in the specification sheet titles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification and applicable specification sheet.
(b) Part number of scrub brush required (see specification sheet).
(c) Selection of applicable level of packaging and packing (see 5.1).

6.3 Supersession data. This specification with Federal Specification Sheets H-B-1490/1 through H-B-1490/8 supersedes the following specifications:

Specification	Date
H-B-1490	September 18, 1969
H-B-444	March 14, 1947
H-B-00444A(GSA-FSS)	November 15, 1968
MIL-B-22784C	31 January 1972
MIL-B-36143B(DSA-DM) (in part)	1 March 1971

The specification sheet and part number applicable to the superseded specifications is indicated in table IV.

TABLE IV. Supersession data cross reference

Federal specification sheet and part numbers	Superseded specification
Brush, Scrub, Hand H-B-1490/1 Part number HB1490/1-P1	Type I, class 1 of H-B-1490
Brush, Scrub, Surgical U-Shaped Block H-B-1490/2 Part number HB1490/2-P1	MIL-B-36143B (DSA-DM) Style B
Brush, Scrub, Household H-B-1490/3 Part number HB1490/3-S1	Type II, class 2 of H-B-1490
HB1490/3-T1	Type II, class 4 of H-B-1490
Brush, Scrub, Clothes H-B-1490/4 Part number HB1490/4-S1	Type III, class 2 of H-B-1490
HB1490/4-PA1	Type III, class 5 of H-B-1490

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TABLE IV. Supersession data cross reference (cont'd)

Federal specification sheet and part numbers	Superseded specification
Brush, Scrub, Floor H-B-1490/5	
Part number HB1490/5-S1	Type IV, class 2 of H-B-1490
-P01	Type IV, class 3 of H-B-1490
-PA1	Type IV, class 5 of H-B-1490
Brush, Scrub, Pot and Pan H-B-1490/6	
Part number HB1490/6-P1	Type I of MIL-B-22784C
Brush, Scrub, General Utility H-B-1490/7	
Part number HB1490/7-P1	Type II of MIL-B-22784C
Brush, Scrub, Pot and Sink H-B-1490/8	
Part number HB1490/8-P1	H-B-00444A (GSA-FSS) and H-B-444

Custodians:

Army - GL
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Review activities:

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